

Lecture 14: MongoDB

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Create, Read, and Delete Operations

Create (Insert)

// Insert a single document

```
db.users.insertOne({  
  name: "Alice",  
  age: 30,  
  email: "alice@example.com"  
});
```

Create, Read, and Delete Operations

Read (Find)

// Find a document

```
db.users.findOne({ name: "Alice" });
```

Delete Operations

Delete

// Delete a single document

```
db.users.deleteOne({ name: "Alice" });
```

Update Operations

Replace vs. \$set

Replace overwrites the entire document, keeping only the `_id`.

```
db.users.replaceOne({ name: "Bob" }, { name: "Robert" });  
// { "_id": ..., "name": "Robert" }  
// (age and email are gone)
```

\$set modifies specific fields, leaving others untouched.

```
db.users.updateOne({ name: "Bob" },  
                   { $set: { name: "Robert" } });  
// { "_id": ..., "name": "Robert", "age": 35, ... }
```

Update: Operators, Upserts, and Multi-updates

Update Operators

Operators like \$inc, \$push, \$pull modify fields.

```
// Increment age by 1
```

```
db.users.updateOne({ name: "Robert" },  
                   { $inc: { age: 1 } });
```

Upserts (Update or Insert)

If no document matches the filter, a new one is created.

```
db.users.updateOne(  
  { name: "Charlie" },  
  { $set: { age: 25 } },  
  { upsert: true }  
);
```

Update: Operators, Upserts, and Multi-updates

Multiple Updates

`updateMany` modifies all documents that match the filter.

```
db.users.updateMany({ city: "New York" },  
                    { $set: { country: "USA" } });
```

Find

Field Conditions

Specify conditions to search (\$gt, \$gte,\$ne).

```
db.users.find({gender: 'm',  
               sid: {$gt: 110}})
```

```
db.users.find({gender: {$ne: 'f'},  
               sid: {$gte: 111}})
```

```
db.users.find({  
  major: {$in: ['CS', 'DSAI']}})
```


Field Conditions

Specify boolean combination of conditions (other than AND).

```
db.users.find({gender: 'f',  
              $or: [{major: 'CS'},  
                   {sid: {$lt: 500}}]  
            })
```

Find: Projections, Ordering

Field Selection (Projection)

Specify which fields to include (1) or exclude (0).

// Return only the name and email fields

```
db.users.find({}, { projection: { name: 1,  
                                email: 1, _id: 0 } }));
```

Ordering (Sort)

Sort documents based on a field. (1 for ascending, -1 for descending).

// Find all users, sort by age in descending order

```
db.users.find({}).sort({ age: -1 });
```

Find: Paging

Paging (skip and limit)

`limit()` restricts the number of results. `skip()` bypasses documents.

// Get the 3rd and 4th youngest users

```
db.users.find({}).sort({ age: 1 }).skip(2).limit(2);
```

Count

Get the total number of documents matching a query.

```
db.users.countDocuments({ age: { $gt: 30 } });
```

Thank You!
Questions?